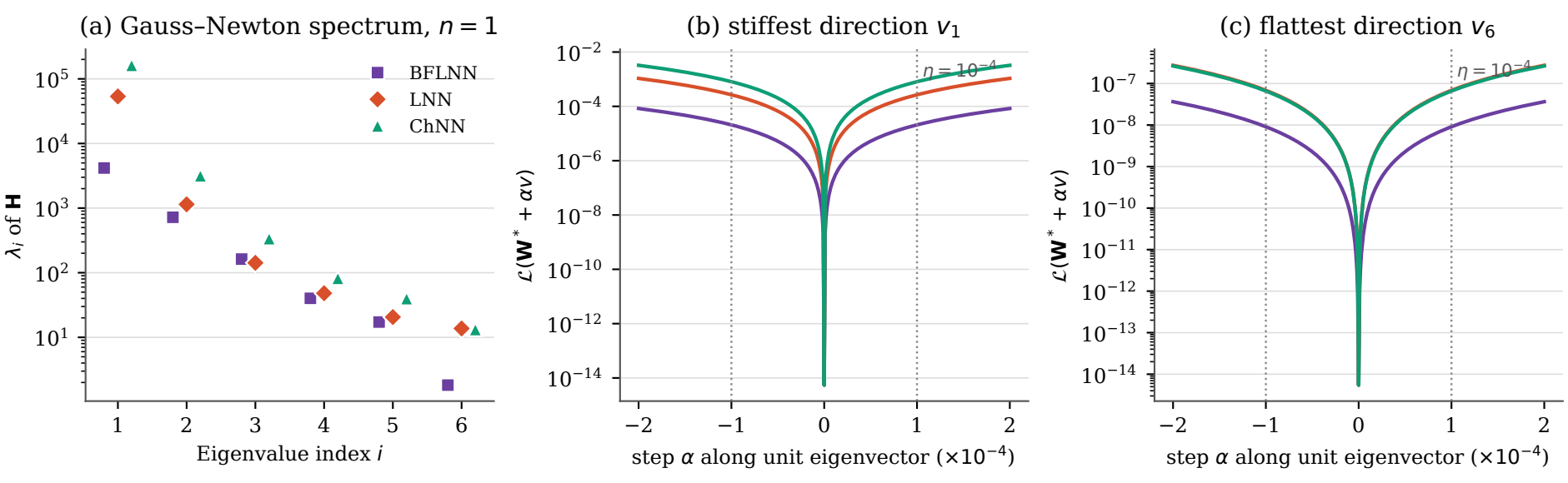

(a) Gauss–Newton spectrum, $n = 1$
BFLNN
LNN
ChNN
λ_i of $\mathbf{H}$
10^5
10^4
10^3
10^2
10^1
1
2
3
4
5
6
Eigenvalue index i
(b) stiffest direction v_1
$\eta = 10^{-4}$
$\mathcal{L}(\mathbf{W}^* + \alpha v)$
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
−2
−1
0
1
2
step α along unit eigenvector ($\times 10^{-4}$)
(c) flattest direction v_6
$\eta = 10^{-4}$
$\mathcal{L}(\mathbf{W}^* + \alpha v)$
10^{-7}
10^{-8}
10^{-9}
10^{-10}
10^{-11}
10^{-12}
10^{-13}
10^{-14}
−2
−1
0
1
2
step α along unit eigenvector ($\times 10^{-4}$)